Md. Abdullah Al Mamun Hridoy

Environmental Researcher | Water Quality Modelling | Hydroclimatic | Machine Learning and IoT

Google Scholar, **Web of Science**, **GitHub**, **LinkedIn**
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PROFESSIONAL SUMMARY

Early-career environmental researcher with interdisciplinary experience in water-quality modelling, aquaculture, environmental pollution assessment, ecological risk analysis, machine learning, deep learning, IoT-based environmental monitoring, hydrology, and sustainable water management. Research activities focus on applying computational and data-driven methods to predict water-quality parameters, assess contaminant dynamics, support environmental decision-making, and improve the management of aquatic and engineered water systems.
Experienced in international multidisciplinary collaboration involving researchers from Europe, Asia, North America, South America, Africa, and Oceania.

RESEARCH INTERESTS

- Environmental Engineering and Sustainable Water Management
- Hydrology and Hydroclimatology
- Water and Wastewater Quality Assessment
- Environmental Pollution and Remediation
- Machine Learning and Deep Learning for Environmental Systems
- IoT-Based Environmental Monitoring and Soft Sensors
- Water Quality Index and Contaminant Prediction
- Microplastic, Macroplastic, and Heavy-Metal Pollution
- Ecological and Human Health Risk Assessment
- Environmental Decision Optimization
- Remote Sensing and Spatial Environmental Analysis
- Nature-Based Solutions and Biochar Technology
- Climate-Resilient Water and Aquaculture Systems

EDUCATION

Bachelor of Science in Fisheries

Faculty of Fisheries, Sylhet Agricultural University, Bangladesh

2022 – Present | Current CGPA: 3.257/4.00 through the 6th semester. Currently enrolled in the seventh semester | Expected completion: July 2027

CURRENT RESEARCH APPOINTMENT

Guest Researcher

UiT The Arctic University of Norway, Tromsø, Norway
June 2025 to Present
abhri9264@uit.no

- Conducting interdisciplinary research at the intersection of environmental engineering, water-quality management, artificial intelligence, and environmental sustainability.
- Collaborating with international researchers on environmental data analysis, computational modelling, manuscript development, and scientific publication.

RESEARCH EXPERIENCE

Water Quality Monitoring and Predictive Modelling

- Developed machine-learning, deep-learning, IoT, and AAH_AquaPredict stacking models for water-quality classification, WQI prediction, and forecasting of dissolved oxygen, TDS, and contamination dynamics.
- Applied ensemble learning, explainable AI, soft sensing, and walk-forward validation to multi-station environmental datasets.
- Developed predictive tools for water-resource management, pollution surveillance, and environmental early-warning systems.

Environmental Pollution and Risk Assessment

- Investigated microplastic, macroplastic, heavy-metal, and hazardous organic contamination in water, soil, sediment, and biological samples.
- Conducted environmental sampling and assessed ecological and human health risks using contamination indices and exposure frameworks.
- Analysed pollutant distribution and aquatic bioindicators to support pollution control and sustainable environmental management.

Water and Wastewater Management

- Modelled water-disinfection decisions by integrating treatment efficacy, disinfection by-product risks, and energy trade-offs.
- Contributed to wastewater surveillance and data-driven management of organic and inorganic contaminants.
- Evaluated interactions among water chemistry, environmental stressors, ecosystem health, and management performance.

Sustainable Environmental Systems

- Applied multi-index frameworks to assess environmental quality, pollution, ecosystem health, and management performance.
- Contributed to research on biochar, sustainable water management, environmental monitoring, and energy-transition systems.
- Integrated field, laboratory, and computational methods to assess environmental resilience and anthropogenic impacts.

PREVIOUS RESEARCH POSITION

Research Assistant

Department of Computer Science and Engineering
Sylhet Agricultural University, Bangladesh
July 2024 to December 2024

Supervisor: Associate Professor Pronojit Kumar Das

- Developed a deep-learning model to predict population trends using environmental and biological datasets.
- Integrated ecological information with artificial intelligence to forecast changes under different environmental scenarios.
- Published research findings contributing to the application of predictive modelling in environmental and aquatic-resource management.

TECHNICAL SKILLS

Programming: Python, R | **Data Science:** Machine learning, deep learning, ensemble modelling, time-series forecasting, and explainable AI | **Environmental Assessment:** Water-quality analysis, WQI, soil and sediment sampling, heavy metals, plastics, and environmental risk assessment | **Geospatial and Monitoring:** QGIS, remote sensing, MaxEnt, and IoT sensors

PUBLICATIONS

Water Quality Modelling and Environmental Monitoring

1. **Hridoy, M. A. A. M.**, Bodini, M., Bordin, C., et al. (2026). Explainable soft-sensing of dissolved oxygen in a multi-station aquatic monitoring network using ensemble machine learning and neural networks. *Results in Engineering*, 112205. https://doi.org/10.1016/j.rineng.2026.112205
2. **Hridoy, M. A. A. M.**, Bodini, M., Mahin, M. R., et al. (2026). Operational early-warning forecasts of aquaculture water quality: Interpretable machine learning for TDS under walk-forward validation. *Ecological Informatics*, 103804. https://doi.org/10.1016/j.ecoinf.2026.103804

3. **Hridoy, M. A. A. M.**, Bordin, C., Pastorino, P., et al. (2026). Neural network-based prediction of water contamination dynamics in a rapidly developing tropical region. *Case Studies in Chemical and Environmental Engineering*, 101342. https://doi.org/10.1016/j.cscee.2026.101342
4. **Hridoy, M. A. A. M.**, Bordin, C., Masood, A., and Masood, K. (2025). Predictive modelling of aquaculture water quality using IoT and advanced machine learning algorithms. *Results in Chemistry, 16*, 102456. https://doi.org/10.1016/j.rechem.2025.102456
5. **Hridoy, M. A. A. M.,** Schneider, P., Bordin, C., Bodini, M., Shawkat, A. I., Maulud, K. N. A., ... & Pastorino, P. (2026). Integrating Advanced Deep Learning with Ecological Indicators for Accurate Prediction of Harmful Algal Bloom Occurrence and Intensity. *Journal of Hazardous Materials Advances*, 101348. https://doi.org/10.1016/j.hazadv.2026.101348
6. **Hridoy, M. A. A. M.,** Pastorino, P., Bordin, C., Bodini, M., Dhar, N., Schneider, P., ... & Maulud, K. N. A. (2026). Operational agricultural water management in river-based aquaculture: A machine-learning approach to predict dissolved oxygen in the Halda River, Bangladesh. *Aquacultural Engineering*, 102693. https://doi.org/10.1016/j.aquaeng.2026.102693
7. **Hridoy, M. A. A. M.,** Pastorino, P., Bordin, C., Bodini, M., Jamee, M. F., Farabi, A., ... & Schneider, P. (2026). Wastewater-based epidemiology of hazardous organics: Multi-biomarker insights and intelligent monitoring technologies. *Journal of Hazardous Materials: Organics*, 100022. https://doi.org/10.1016/j.hazmo.2026.100022
8. **Hridoy, M. A. A. M.**, Shawkat, A. I., Bordin, C., et al. (2025). Advanced machine learning models for accurate water quality classification and WQI prediction: Implications for aquatic disease risk management. *Science of the Total Environment, 1008*, 180965. https://doi.org/10.1016/j.scitotenv.2025.180965
9. **Hridoy, M. A. A. M.**, Ditthakit, P., Hosen, A., et al. (2026). Machine learning-driven prediction of heavy metal pollution for aquatic health surveillance and disease control. *iScience, 29*(7). https://doi.org/10.1016/j.isci.2026.116549
10. **Hridoy, M. A. A. M.,** Paul, P. B., & Masood, A. (2025). Monitoring flood-prone urban areas of Sylhet, Bangladesh through water quality remote sensing. *Discover Environment*, *3*(1), 125. https://doi.org/10.1007/s44274-025-00322-7
11. **Hridoy, M. A. A. M.**, Akter, P., Acharjee, M. R., et al. (2026). Decision optimization for sustainable water disinfection: Computational modelling of DBP risk, efficacy, and energy trade-offs. *Case Studies in Chemical and Environmental Engineering*, 101430. https://doi.org/10.1016/j.cscee.2026.101430
12. **Hridoy, M. A. A. M.**, Sagor, P. S., Akter, P., et al. (2026). Decision optimization for inorganic contaminants in water systems using ecological informatics for sustainable management. *Case Studies in Chemical and Environmental Engineering*, 101362. https://doi.org/10.1016/j.cscee.2026.101362
13. Pinthong, S., Salaeh, N., Pham, Q. B., **Hridoy, M. A. A. M.,** Bordin, C., & Ditthakit, P. (2026). Coupled Spiking Neural Network–Extra Trees Framework for Daily Streamflow Imputation with ERA5-Land Integration. *Engineered Science*, *41*, 2204. http://dx.doi.org/10.30919/es2204
14. Salaeh, N., Pinthong, S., Pham, Q. B., **Hridoy, M. A. A. M.,** Bordin, C., & Ditthakit, P. (2026). Adaptive Parameter-Switching GR2M Model for Streamflow Simulation in Tropical Basins of Southern Thailand. *Engineered Science*, *41*, 2205. http://dx.doi.org/10.30919/es2205

SPECIALIZED TRAINING AND INTERNATIONAL SUMMER SCHOOLS

WATER4MED Summer School on Water Quality and Groundwater Modelling

Polytechnic University of Bari, Bari, Italy
8–11 June 2026

- Selected participant in the WATER4MED Summer School focused on **water-quality assessment and groundwater modelling**.
- Program hosted by the **Department of Civil, Environmental, Land, Building Engineering and Chemistry**.
- Received specialized training relevant to **civil and environmental engineering, groundwater systems, water-resource management, and environmental modelling**.
- **Process-Based Groundwater Modelling:** Hands-on MODFLOW implementation through the Water4MED Summer School, including model setup, parameterization, boundary conditions, simulation, and output interpretation

Geneva Summer School: Microplastics, from Environmental Impact to Policy, Innovation, and Public Awareness

University of Geneva, Geneva, Switzerland
15–20 June 2026

- Awarded a **fully funded scholarship** and participated in person in the Geneva Summer School 2026.
- Completed interdisciplinary training on **microplastic pollution, environmental impacts, policy responses, technological innovation, and public awareness**.
- Strengthened expertise in translating environmental contamination research into **policy, management, and sustainable engineering solutions**.

- Nominated for the **USREN Prize** in recognition of potential as a promising young researcher.

CONFERENCES AND SCIENTIFIC PRESENTATIONS

✧ 2nd International Conference & Exhibition, **ORAL presenter,** Science City, Kolkata, India (Session Best Presenter) [2024]

- ✧ International Symposium on Marine Litter and Coastal Ecosystem Nexus (MariNEX), **POSTER presenter**, KU, Bangladesh [2025]
- ✧ 2nd Asian Student Council Symposium (ASCS2023), ISCB- International Society for Computational Biology, USA. **Online participant** [2023]
- ✧ 2nd International Conference on Sustainable Fisheries (ICSF) 2022, Faculty of Fisheries, Sylhet Agricultural University, Sylhet-3100. **POSTER presenter** [2022]
- ✧ 3rd Biennial International Conference 2023, Fisheries Society of Bangladesh. **ORAL and POSTER presenter** [2024]
- ✧ International Conference on Advanced Agricultural Research, Sylhet Agricultural University Research System (SAURES), Sylhet-3100. **POSTER presenter** [2024]
- ✧ 3rd International Conference on Advances of Life Science, **ORAL presente**r (Session Best Presenter), MBSTU, Bangladesh. [2025]
- ✧ 38th BSM Annual International Conference, **ORAL presenter**, Brac University, Bangladesh [2024] (Session Best Presenter)
- ✧ 1st International Scientific Conference on Sustainable Aquaculture & Fisheries, **ORAL presenter,** CVASU, Bangladesh [2025]

REVIEWER EXPERIENCE

Evidence and reviewer certificates:
Reviewer Experience Portfolio

REFERENCES

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